ModelIt!

Teacher's Guide

Your Music Is a Wave

How Sound, Light, and WiFi All Follow the Same Rules

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-PS4-1, MS-PS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-PS4-1, MS-PS4-2

Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave. Develop and use a model to describe that waves are reflected, absorbed, or transmitted through various materials.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model describing how amplitude and frequency affect wave energy and perception.
Disciplinary Core Idea	**PS4.A Wave Properties** — A simple wave has a repeating pattern with specific amplitude, frequency, and wavelength related to energy.
Crosscutting Concept	**Patterns** — Students identify the repeating pattern structure of waves and how changing pattern parameters changes energy and perception.

CAST Testing Alignment

MULTIPLE CHOICE:

A sound wave has its amplitude doubled while frequency stays the same. What happens to the energy of the wave?

CONSTRUCTED RESPONSE:

Using wave properties, explain why a concert sounds different in a carpeted room versus a room with all hard surfaces. Include the concepts of reflection and absorption.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every sound you hear, every color you see, every WiFi signal your phone receives — they're all waves. Understanding wave properties reveals the physics behind music, communication, and even medical imaging.

Wave Basics

A wave is a repeating disturbance that transfers energy without transferring matter. The key properties are: amplitude (height from baseline), frequency (cycles per second, measured in Hertz), wavelength (distance between identical points), and speed (depends on the medium).

Amplitude and Energy

Wave energy is proportional to amplitude SQUARED. This means doubling the amplitude quadruples the energy. In sound, amplitude determines loudness. In light, amplitude determines brightness. This squared relationship is why a small increase in volume requires a large increase in speaker power.

Frequency and Perception

Humans can hear frequencies from about 20 Hz (deep bass) to 20,000 Hz (high pitch). Frequency determines the pitch of sound and the color of light. Musical notes correspond to specific frequencies: middle C is 262 Hz, the A above it is 440 Hz.

Wave Interactions with Materials

When waves encounter a new material, they can be: reflected (bounced back — echo), absorbed (converted to heat — sound dampening), or transmitted (passed through). Most materials do all three to varying degrees. Hard surfaces reflect more sound; soft surfaces absorb more.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify wave amplitude, wave frequency, wave energy, and sound perception as system components.

E — Establish Relationships
Students determine that amplitude strongly increases energy (squared relationship), and frequency affects pitch perception.

V — Visualize & Model
Students build the wave model showing how properties connect to energy and perception.

E — Evaluate Outcomes
Students compare whisper, screaming high note, and bass drop scenarios.

R — Revise & Extend
Students add medium density or reflection to model material interactions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Right now, invisible waves are passing through you. Sound, WiFi, radio — they're everywhere."*

Watch For

Play a short clip of music. Ask: How did that sound get from the speaker to your ears?

Slide 2: Learning Objectives

3 min

Say: *"Today we decode the physics of music, sound, and all waves."*

Watch For

Pre-teach amplitude (loudness) and frequency (pitch) with hand waves.

Slide 3: Big Question

3 min

Say: *"Your favorite song is just air molecules vibrating in a pattern. How?"*

Watch For

Demo: speaker with visible cone vibration.

Slide 4: LEVER Framework

2 min

Say: *"We'll model the wave properties that make music possible."*

Watch For

Preview the LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What properties define a wave? What can we control?"*

Watch For

Sort components. Use a slinky or rope to demonstrate.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the surprise: doubling amplitude QUADRUPLES energy. Not doubles — quadruples!"*

Watch For

Map relationships. Emphasize the squared relationship.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's hear the difference between a whisper and a scream in wave terms."*

Watch For

Students adjust amplitude and frequency, observe energy changes.

Slide 8: Discoveries

5 min

Say: *"ALL waves follow these same rules — sound, light, WiFi, ocean waves."*

Watch For

Connect to real-world wave applications.

Slide 9: STEM Challenge

12 min

Say: *"A music school hired YOU to design their performance hall. Acoustics matter!"*

Watch For

Teams design venues considering reflection, absorption, and room shape.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Wave Amplitude, Wave Frequency
- **INTERNAL:** Wave Energy, Sound Perception

Relationships

- **Wave Amplitude to Wave Energy:** POSITIVE (+) — Higher amplitude means more energy — in fact, energy scales with amplitude SQUARED (double amplitude = 4x energy).
- **Wave Frequency to Sound Perception:** POSITIVE (+) — Higher frequency waves are perceived as higher pitch. Lower frequency = lower pitch (bass).
- **Wave Energy to Sound Perception:** POSITIVE (+) — More wave energy is perceived as louder sound. The wave carries more energy to your eardrum.

Key Discoveries

- Amplitude controls LOUDNESS — bigger waves carry more energy
- Frequency controls PITCH — more cycles per second = higher pitch
- Doubling the amplitude QUADRUPLES the energy (squared relationship!)
- Sound, light, radio, and WiFi are ALL waves following the same basic rules

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Sound can travel through empty space"	Sound requires a medium (matter) to travel through. Sound travels through air, water, and solids, but NOT through a vacuum. In space, no one can hear you scream!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Loud sounds travel faster"	Volume (amplitude) does NOT affect the speed of sound. A whisper and a scream travel at the same speed (~343 m/s in air). Amplitude only affects energy, not speed.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Higher frequency means more energy"	Frequency affects pitch, not necessarily total energy. Amplitude is the primary driver of wave energy. A loud bass note (high amplitude, low frequency) can carry more energy than a quiet high note (low amplitude, high frequency).